

An Auxiliary Finite Lattice Model Inspired by SIT

Gauge Covariance, Structural Orbit Embedding, Fixed-Magnitude Factorization, Quartic Pinning, and Open Transfer

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Abstract

The first public version of this paper described a finite two-dimensional $U(1)$ gauge-scalar system as a lattice specification of Super Information Theory (SIT). That framing exceeded what its written action and source record established. The deposited link transformation was also reversed for the displayed kinetic term, and the proposed toy-sector embedding preserved configurations, gauge orbits, and holonomy observables but not the action or Gibbs weights. This Version 2 retains the legitimate mathematical program and repairs those defects without silently replacing the historical record. It defines the system as an auxiliary finite lattice model inspired by SIT, derives the correct gauge action and an invariant edge variable, proves a structural orbit correspondence on a finite simply connected square box, exhibits the nonconstant edge functional that prevents action-level equivalence, proves fixed-magnitude factorization of a positive scalar constraint for phase / link events, and derives explicit quartic magnitude and counting bounds. It also restores the original window taxonomy, finite-to-finite transfer program, falsification route, dimensional boundary, and limit-ordering questions. Deterministic checks reproduce the corrected gauge invariance, reject the deposited inverse link law on a fixed fixture, verify the edge-action mismatch, evaluate the half-line Gaussian, and test the quartic bounds. Transfer of toy-model abundance constants remains open. The model does not specify physical SIT, establish a continuum or thermodynamic limit, validate SIT empirically, prove an HSIT or SIT / Riemann theorem, or prove the Riemann Hypothesis.

Scope and evidence boundary

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1. The Problem Before the Name

Suppose a researcher has a proposed physical theory and wants to study a finite lattice analogue. Before the analogue can support a theorem, a reader must know what lives on the lattice, how those objects transform, how they are weighted, which measurements count as observables, and which limiting process would be needed to return to a continuum theory. Similar vocabulary is not enough. A finite model can be mathematically coherent while failing to be a discretization of the physical theory that motivated it.

The established lattice-gauge account begins with a finite graph or cell complex, fields on vertices or edges, a reference measure, an action, a gauge group, and declared observables. Gauge covariance is checked term by term. Finite-volume probability statements depend on the measure and action actually written. A continuum interpretation requires a separate scaling and convergence argument; it does not follow from a suggestive resemblance.

The first public version of this paper contained a useful finite model but crossed those levels. It treated four unresolved SIT source slots as if they could be filled later without changing the model, called a structural restricted-sector correspondence an exact embedding, and used a link transformation that does not preserve its own kinetic term. The repair is to perform four operations in order:

1. freeze the historical paper and identify every changed claim;
2. define the finite mathematical object independently of a physical interpretation;
3. prove only what follows from that object and its chosen measure; and
4. state any connection to SIT as motivation or as a future transfer problem.

Only after those operations are understood is the model named here: **an auxiliary finite lattice model inspired by SIT**. “Auxiliary” means that the model can organize and test a bounded mathematical question without being identified with physical SIT.

This classification has concrete failure conditions. It would have to be revised if the gauge calculation below failed, if the orbit map were not a bijection in the stated geometry, if the scalar integral did not factorize at fixed magnitude, or if the quartic estimate failed. Conversely, none of those finite results would by itself establish a faithful SIT discretization. That stronger claim would require a public, version-stable SIT action and a term-by-term discretization and convergence theorem.

2. Version Lineage and Source-Faithful Correction

Zenodo record 21231885 remains the immutable Version 1 baseline. Version 2 does not backdate its corrected terminology or imply that the deposited PDF already contained the present distinctions. The correction preserves four separately auditable layers:

- the original deposited wording and PDF;
- the strongest supported intent, namely a finite lattice route for exploring SIT-inspired field and constraint ideas;
- the external mathematical state supplied by lattice gauge and gauge-scalar theory; and
- the surviving Micah-authored proposition: this chosen finite model has structural, factorization, and pinning results, while transfer to any fuller theory remains an open problem.

The material correction differential is:

ID	Deposited position	Version 2 disposition
C01	The model was described as a lattice specification of SIT.	Reclassified as an auxiliary finite two-dimensional model inspired by SIT.
C02	The link law used the inverse orientation for the displayed kinetic term.	Replaced by the gauge-covariant orientation derived in Section 4.
C03	The restricted sector was called an exact toy embedding.	Restricted to a structural bijection of configurations modulo gauge and preservation of holonomy observables.
C04	The toy and auxiliary actions were treated as sufficiently identical for lifted conclusions.	The nonconstant edge functional is explicit; action, Gibbs-weight, energy-ordering, and spectral transfer are not inherited.
C05	A nondegenerate minimum was treated as sufficient for general pinning.	The unconditional result is limited to the displayed quartic; a generalization requires a unique zero and a positive ratio bound.

ID	Deposited position	Version 2 disposition
C06	The radial reference measure and normalizability route were under-specified.	The deliberate ordinary radial measure and finite-volume integrability argument are stated.
C07	Scalar / holonomy factorization could be read as unconditional.	It is conditional on a fixed radial profile and phase / link events; total-action windows obey a convolution.
C08	The lift theorem hid the conclusion-bearing transfer premise.	It is now a conditional dependency statement, and the transfer theorem is explicitly open.
C09	The lattice current depended on an unresolved SIT source slot.	It is derived only as a current of the chosen auxiliary action, with no physical electromagnetic interpretation.
C10	Later companion or proof-program proximity could be read as supporting the lift.	No companion title, finite proof packet, or later program supplies the missing transfer theorem.

The correction is additive rather than destructive. Version 2 retains the finite lattice geometry, phase and link variables, positive scalar, binned mode ladder, excitation and holonomy windows, fixed-magnitude factorization, half-line Gaussian, quartic pinning, transfer constants, schedule question, falsification proposal, dimension problem, and limit-ordering problem. What it removes are promotions not supported by those objects.

3. Finite Object, Fields, Measure, and Action

Let $\Lambda_n = \{0, \dots, n\}^2$ be a finite simply connected square box. Denote its vertices, positively oriented nearest-neighbor edges, and plaquettes by V , E , and P . For an edge $e = (x \rightarrow y)$, the fields are

$$\psi_x = R_x \exp(i \theta_x), \quad R_x \geq 0; \quad u_{xy} \in U(1); \quad \rho_x > 0. \quad [1]$$

The phase θ_x belongs to a vertex field, u_{xy} is a compact internal link variable, and ρ_x is a positive scalar. The parameters κ , β , λ , and μ are strictly positive. Let $F: [0, \infty) \rightarrow (0, \infty)$ be continuous.

The reference measure uses normalized Haar measure for every vertex phase and link, ordinary dR_x on $[0, \infty)$, and $d\rho_x$ on $(0, \infty)$:

$$d\nu = \prod_x [d\theta_x / (2\pi) dR_x d\rho_x] \prod_e [du_e]. \quad [2]$$

The radial choice is deliberate. It is not the planar Lebesgue measure $R_x dR_x d\theta_x$ induced from the complex plane. Changing the measure changes the model and may change probability statements, even if the set of configurations is unchanged.

The auxiliary action is

$$S = \kappa \sum_{(x \rightarrow y)} |\psi_x - u_{xy} \psi_y|^2 + \beta \sum_p (1 - \operatorname{Re} u_{\text{boundary}(p)}) + \sum_x [\lambda (R_x^2 - 1)^2 + \mu (\rho_x - F(R_x^2))^2]. \quad [3]$$

Here $u_{\text{boundary}(p)}$ is the oriented product of link variables around a plaquette. The pinning radius, quartic potential, positive-scalar penalty, two-dimensional geometry, and measure are definitions of this auxiliary model. They are not claimed to be terms recovered verbatim from SIT.

3.1 Finite-volume existence

Every term in the action is nonnegative. The phase and link domains are compact. For each fixed radius, the positive-scalar integral is a half-line Gaussian bounded above by $\sqrt{\pi/\mu}$. The quartic term grows as λR_x^4 , so its ordinary radial integral is finite. Because the lattice has finitely many vertices and edges, the partition function

$$Z = \int \exp(-S) d\mu \quad [4]$$

exists, is finite, and is positive. This is a finite-volume statement for the chosen model. It is not an infinite-volume Gibbs construction.

3.2 Binned phase ladder, observables, and windows

The original transfer program used a finite phase ladder so that its combinatorial statements were literal counts rather than integrals over a continuous group. Preserve that construction as follows. Choose an increasing integer sequence f_j . At stage j , restrict both the vertex phases and the link phases to μ_{f_j} , the group of f_j -th roots of unity. Binning only the links is insufficient: the invariant edge contains the difference of two vertex phases, so both phase families must lie in the same finite group if the result is to remain in that group. The radial and positive-scalar variables remain continuous unless a separately declared approximation bins them.

Choose an observed plaquette set P_{obs} , equal to all plaquettes unless otherwise stated. For an invariant edge assignment ω , write $\sigma_p(\omega)$ for the oriented product around plaquette p . Its excitation count is the number of observed plaquettes for which this product is not the identity. An excitation-count window bounds that integer. A holonomy window is any event measurable from the declared set of plaquette products. A total-action window instead depends on the full configuration.

On a finite phase group, the Wilson plaquette energy of every nontrivial observed plaquette is bounded below by $\beta[1 - \cos(2\pi/f_j)]$. Consequently, an excitation count supplies a lower bound on the corresponding observed plaquette energy. The converse requires care because an energy budget can be distributed across plaquettes in different ways. This comparison connects a combinatorial window to one part of the action; it does not eliminate the edge functional, radial energy, or positive-scalar energy.

The binned radial observable from Version 1 can also be retained without promoting it. For a declared width δ_j , map every R_x to its radial bin. This produces a finite descriptive profile at stage j . It does not imply that a radial continuum limit exists or that the profile is a physical SIT observable.

4. Gauge Covariance, Invariant Edges, and the Auxiliary Current

For any $g: V \rightarrow U(1)$, the transformation compatible with the kinetic term is

$$\psi_x \rightarrow g_x \psi_x; u_{xy} \rightarrow g_x u_{xy} \text{conjugate}(g_y); \rho_x \rightarrow \rho_x. \quad [5]$$

Indeed, $g_x \psi_x (g_x u_{xy} \text{conjugate}(g_y)) (g_y \psi_y)$ equals $g_x (\psi_x u_{xy} \psi_y)$, so the kinetic modulus is unchanged. The vertex factors telescope around every plaquette, and the on-site terms depend only on gauge-invariant magnitudes. The full action is therefore gauge invariant.

The deposited transformation used $u_{xy} \rightarrow u_{xy} g_y \text{conjugate}(g_x)$. For generic phases it leaves a residual vertex factor and fails to preserve the displayed kinetic term. The supplementary negative regression fixes one nontrivial configuration and verifies that the deposited law changes the action while the corrected law does not.

When both endpoint magnitudes are nonzero, define

$$\omega_{xy} = \frac{\text{conjugate}(\psi_x) u_{xy} \psi_y}{R_x R_y}. \quad [6]$$

Substitution shows that ω_{xy} is gauge invariant. Products of these variables around a plaquette equal the corresponding link holonomy because the vertex phases cancel.

Varying the chosen action with respect to a vertex phase gives the oriented edge current, up to the declared orientation convention,

$$J_{xy} = 2\kappa \text{Im}(\text{conjugate}(\psi_x) u_{xy} \psi_y). \quad [7]$$

At a stationary point the signed discrete divergence of this current vanishes because the on-site terms are phase independent. This is a mathematical current of the auxiliary action. It is not identified here with physical electromagnetism or with a sourced SIT current.

5. Structural Restricted-Sector Correspondence

Fix an integer $f \geq 2$ and let μ_f be the group of f -th roots of unity. Restrict the model to

$$R_x = 1; \rho_x = F(1); \exp(i\theta_x) \in \mu_f; u_{xy} \in \mu_f. \quad [8]$$

In this sector, every invariant edge ω_{xy} belongs to μ_f .

Proposition 1 (structural orbit correspondence). On a finite simply connected square box, the map from restricted phase / link configurations modulo the corrected gauge action to edge assignments ω in $(\mu_f)^E$ is a bijection. It preserves plaquette holonomies and all observables depending only on those holonomies.

Proof. Gauge invariance follows from Equation 6. For surjectivity, choose all vertex phases equal to one and set each link equal to the desired edge assignment. If two restricted configurations produce the same ω , the ratio of their vertex phases defines a gauge function relating their links. Plaquette products are preserved because vertex factors telescope. This proves the stated correspondence.

The proposition is structural, not an action identity. At unit magnitude,

$$\kappa |\psi_x u_{xy} \psi_y|^2 = 2\kappa (1 - \text{Re } \omega_{xy}). \quad [9]$$

Therefore the auxiliary action differs from the plaquette-only toy action by

$$\Delta S_{\text{edge}} = 2\kappa \sum_e (1 - \text{Re } \omega_e). \quad [10]$$

This functional is nonnegative but generally nonzero and nonconstant. Consequently, the proposition preserves configuration classes, gauge orbits, plaquette holonomies, and holonomy-defined windows. It does not preserve Boltzmann weights, energy orderings, action windows, partition functions, or spectra.

5.1 Window taxonomy

The distinction becomes precise through three window classes:

Window	Defining information	Structural correspondence	Scalar factorization at fixed R
Excitation-count window	Number of nontrivial observed plaquette holonomies	Yes	Yes
Holonomy window	Any event measurable from declared plaquette holonomies	Yes	Yes

Window	Defining information	Structural correspondence	Scalar factorization at fixed R
Total-action window	Full action bounded by K, optionally with a radial cutoff	No, because of the edge functional	No; convolution only

This table preserves an important aim of Version 1 while changing its logical consequence. A holonomy-counting result may be compared across the structural map after all hypotheses are checked. A result whose proof consumes weights, energies, action budgets, or spectra needs a new comparison theorem.

6. Fixed-Magnitude Factorization

Fix a radial profile R. Let W be an event depending only on vertex phases and links. Write

$$S = S_{\text{phase}}(\theta, u; R) + \sum_x [\lambda(R_x^2 - 1)^2 + \mu(\rho_x - F(R_x^2))^2]. \quad [11]$$

Define the local positive-scalar factor

$$z_x(R) = \int_0^\infty \exp[-\mu(\rho - F(R^2))^2] d\rho = \sqrt{\pi/\mu} [1 + \text{erf}(\sqrt{\mu} F(R^2))] / 2. \quad [12]$$

Theorem 2 (fixed-magnitude factorization). Under the stated reference measure,

$$Z(W; R) = \exp[-\lambda \sum_x (R_x^2 - 1)^2] \prod_x z_x(R_x) Z_{\text{phase}}(W; R). \quad [13]$$

Proof. At fixed R, the indicator of W and the phase / link action are independent of ρ . The remaining integrand is nonnegative and factors into finitely many half-line Gaussians and a compact phase / link factor. Tonelli's theorem permits the iterated product, producing Equation 13.

The theorem implies conditional independence of the positive-scalar field and holonomy data under the Gibbs measure at a fixed radial profile. It does not imply unconditional independence after integrating over R, because both sectors can then be statistically coupled through the shared radii.

The local factor is always positive and no larger than $\sqrt{\pi/\mu}$. If $F(R^2)$ stays between two positive constants throughout a declared pinning band, the error-function formula supplies uniform positive lower and upper bounds there. These bounds show that the positive-scalar integration cannot reweight one holonomy sector relative to another at the same fixed radial profile. They still do not compare two different radial profiles.

For a total-action event, the scalar and phase energies compete inside one budget. The exact relation has the form

$$Z(S \leq K; R) = \int d\rho \exp[-E_\rho(\rho; R)] Z_{\text{phase}}(S_{\text{phase}} \leq K - E_\rho(\rho; R); R), \quad [14]$$

with the phase factor set to zero when the remaining budget is negative. This is an average over changing energy budgets - a convolution - not a product.

7. Quartic Magnitude Pinning

Lemma 3 (quartic pinning). Every configuration satisfying $S \leq K$ obeys

$$\sum_x (R_x^2 - 1)^2 \leq K/\lambda; \quad \sum_x (R_x - 1)^2 \leq K/\lambda. \quad [15]$$

Proof. Nonnegativity of all action terms bounds the quartic contribution by K. Since $R_x \geq 0$, $(R_x^2 - 1)^2 = (R_x - 1)^2 (R_x + 1)^2$ and $(R_x + 1)^2 \geq 1$. Both inequalities follow.

For every $\delta > 0$, the number of vertices outside the pinning band satisfies

$$\#\{x: |R_x - 1| > \delta\} \leq K/(\lambda \delta^2). \quad [16]$$

This is a deterministic action-window estimate; it is not a statement that typical configurations concentrate in a thermodynamic limit.

For a general nonnegative potential $U(R^2)$, a comparable result on a bounded radial interval requires more than a nondegenerate local minimum. The pinning radius must be the unique zero on that interval, and the ratio $U(r^2)/(r-1)^2$ must have a strictly positive lower bound away from and near the pinning point. Finite-volume normalizability also requires sufficient growth for the declared radial measure. These requirements prevent an unnoticed second zero or weak tail from invalidating the estimate.

8. The Open Toy-to-Auxiliary Transfer Program

Version 1 organized a transfer program around seven quantities from the companion finite toy-model architecture. Version 2 preserves that research question but does not assume its answer.

Quantity	Intended role in a future transfer theorem
Surplus	Per-volume growth exponent in an abundance count
Net growth	Retained-sector growth across refinement stages
Unit translation	Constant controlling a unit-holonomy translation step
Stable growth	Uniform growth across stage carriers
Splitting	Orbit-splitting multiplicity under refinement
Multiplicity	Gauge-orbit-size contribution to counts
Extension	Continuation of window-compatible configurations

The table is a research inventory, not a theorem statement. Each row needs a version-stable definition from the companion toy-model source before a transfer proof can consume it. In particular, two quantities with similar names are not interchangeable merely because both describe growth.

Let f_j increase while radial and scalar tolerances δ_j and ϵ_j decrease. A candidate pinned window requires every radius to lie within δ_j of one and every positive scalar to lie within ϵ_j of its local center. The simplest illustrative schedule sets both tolerances proportional to $1/f_j$, but no theorem here selects that rate.

The central open question is whether each declared toy quantity transfers to the auxiliary model with controlled loss at fixed finite volume, after the nonconstant edge functional and radial fluctuations are restored. A valid transfer theorem must state:

1. the exact companion theorem and every hypothesis being transferred;
2. whether the target is a count, a probability, a partition function, or an action-window statement;
3. how the edge functional changes the target;
4. how errors depend on f_j , δ_j , and ϵ_j ;
5. whether the estimate is uniform in box size;
6. the order of refinement, pinning, volume, and any lattice-spacing limits;
7. the role of topology and dimension; and
8. a counterexample or failure condition when those controls do not hold.

Conditional corollary. If a separately proved theorem transfers the required constants with their hypotheses and errors controlled, then the matching finite toy conclusion follows for the declared auxiliary windows. This statement exposes the dependency; it does not prove the transfer premise.

8.1 Falsification route

The finite claim is computationally attackable. On small lattices one can measure a declared quantity under a sequence of narrowing pinning windows and compare it with the toy value. A discrepancy that remains outside the proposed error bound as the tolerances shrink refutes that proposed transfer statement. Agreement supplies numerical evidence for the finite conjecture but is not a proof and does not establish a continuum theory.

Three negative controls are essential. First, the deposited inverse link law must fail on a nontrivial fixture; otherwise the regression is too weak to distinguish the corrected convention. Second, at least two restricted configurations with different edge functionals must be exhibited; otherwise a constant-action fixture could conceal the failure of Gibbs equivalence. Third, the fixed-magnitude product formula must be contrasted with a total-action window whose remaining phase budget changes with ρ ; otherwise the intended boundary between factorization and convolution has not been tested.

8.2 Dimension and limit order

The structural orbit proof above is stated for finite simply connected two-dimensional boxes. Higher-dimensional plaquette data obey additional constraints, and plaquette holonomy alone need not provide the same complete orbit description. The thermodynamic limit requires estimates uniform in volume. A continuum limit additionally requires a lattice-spacing map, renormalized parameters, and convergence of observables. None of those limits is supplied here.

9. Relationship to SIT and to Lattice Gauge-Scalar Theory

The historical motivation came from SIT's attempt to relate coherence and time-density concepts. The public SIT Version 79 record is a broad continuum physical proposal; it does not provide a term-by-term source for Equation 3. Subsequent private revision work is not used as a public source. Accordingly, the auxiliary link is an internal mathematical variable, not ordinary electromagnetism; the positive scalar is a model field, not a measured local time density; and the quartic pinning radius is a chosen model feature, not a validated physical vacuum.

The established comparison is narrower and useful. Compact links, plaquette holonomies, gauge-covariant scalar hopping, and frozen-magnitude sectors belong to the lattice gauge and gauge-Higgs traditions. Wilson introduced the plaquette-action framework for lattice gauge fields. Osterwalder and Seiler, Fradkin and Shenker, and Kennedy and King developed rigorous results for related lattice gauge-scalar systems under their respective assumptions. Those sources provide mathematical context. They do not prove this paper's model-specific orbit map, scalar factorization, pinning bound, or transfer conjecture.

A future claim of faithful SIT discretization would require a separate project: freeze a public physical action; map every continuum field and symmetry to lattice variables; identify physical and auxiliary gauge sectors; state the reference measure; choose the dimension and boundary conditions; derive the scaling dictionary; and prove term-by-term convergence of declared observables. The current finite results cannot substitute for that work.

10. Reproducibility and Computational Checks

The accompanying code evaluates deterministic consequences of the displayed finite model. Its fixed fixture and policy checks cover:

Check	Result represented by the package
Corrected gauge law	Action preserved to numerical tolerance
Deposited inverse link law	Fails on the fixed nontrivial fixture
Invariant edge	Preserved under the corrected transformation
Plaquette holonomy	Preserved under the corrected transformation
Edge-action mismatch	Exact formula reproduced and shown nonconstant
Half-line Gaussian	Closed form matches deterministic quadrature
Quartic pinning	Magnitude and counting inequalities pass the fixture
Claim boundary	Open transfer and forbidden promotions remain explicit
Optimization safety	No correctness-bearing Python assertions disappear under optimization
Path privacy	Public release materials contain no machine-local source paths

The frozen gauge fixture gives an action of approximately 9.3165352352 before and after the corrected transformation. Under the deposited inverse orientation it gives approximately 24.3442324194, an absolute change of about 15.0276971842. The invariant-edge error is below 5×10^{-16} , and the plaquette-holonomy error is below 2×10^{-16} . For the nontrivial unit-radius fixture, the directly evaluated kinetic contribution and the edge formula both give approximately 2.6568792374, while trivial edges give zero. The closed-form half-line Gaussian and deterministic quadrature differ by less than 5×10^{-16} on the declared test input.

The quartic fixture uses four radii and a positive additional action contribution. Its quartic sum is approximately 1.3753, its squared radial distance sum is approximately 0.29, and both remain below the bound implied by the total action and coupling. These numbers are regression fixtures, not estimates of physical parameters or evidence about typical lattice states.

The checker is a regression instrument, not a theorem prover. A passing result does not prove the structural proposition for arbitrary lattices, establish a uniform transfer estimate, construct an infinite-volume Gibbs measure, prove a continuum limit, or validate a physical interpretation.

The exact local run passes all twelve governed checks and all nine unit tests, including the required negative regression. The optimized and ordinary runs produce the same governed result. This is same-host deterministic reproduction, not independent specialist review or cross-platform reproduction.

11. Claim Status

Claim	Version 2 status
Finite-volume partition function for the displayed measure and action	Derived
Corrected finite gauge invariance	Derived and computationally checked
Deposited inverse link orientation	Rejected by derivation and negative regression
Structural restricted-sector orbit correspondence	Proved for the stated finite 2D geometry
Action-level or Gibbs-level exact toy embedding	Retracted
Fixed-R scalar / holonomy factorization	Proved for phase / link events
Unconditional scalar / holonomy independence	Not claimed
Total-action product factorization	Not claimed; convolution retained
Quartic pinning and counting bounds	Proved for the displayed quartic
General-potential pinning	Conditional on explicit unique-zero, ratio, and growth assumptions
Toy-to-auxiliary constant transfer	Open
Uniform thermodynamic or continuum lift	Open
Faithful SIT discretization	Not supplied

Claim	Version 2 status
Physical SIT, full SIT, HSIT, SIT / Riemann, or RH theorem	Not proved and outside scope

12. Discussion

The correction makes the paper more useful, not merely narrower. The original paper’s central instinct was that a finite model should separate what is already calculable from what remains conjectural. Version 2 realizes that instinct more cleanly. The model is fully declared; the gauge law is correct; the orbit correspondence is proved at the level it actually preserves; the action mismatch is visible; the scalar constraint has an exact conditional factorization; and the quartic term yields explicit bounds.

The remaining transfer question is now sharper. Structural equivalence does not automatically control weighted statistics. The missing theorem must bridge that gap by quantifying the effect of the edge functional and radial fluctuations on the specific quantity being transferred. This reframing replaces an ambiguous “lift” with a family of finite, refutable comparison problems.

The result also clarifies how SIT-inspired mathematical exploration can proceed responsibly. A physical theory may motivate an auxiliary object, but the auxiliary object earns its conclusions from its own definitions and proofs. If a later source-stable SIT discretization is built, the present model can serve as a comparison case or a controlled counterexample. It cannot be declared identical in advance.

This division also protects the original creative aim. The paper did not need the model to be physical SIT in order to ask whether a coherence-like phase sector and a positive scalar constraint can coexist without changing a declared class of holonomy counts. The fixed-magnitude theorem answers that bounded question affirmatively for the chosen action. The answer becomes false or at least unproved when the radial profile is integrated out, the event consumes the total action, or a physical interpretation is added. Those boundaries turn one broad claim into several precise research questions rather than discarding the underlying program.

13. Limitations and Open Problems

The following work remains open:

1. prove or refute a fixed-volume transfer theorem for at least one declared counting quantity;
2. quantify the effect of the nonconstant edge functional on weighted windows;
3. identify schedules for which radial and scalar pinning errors vanish;
4. produce a uniform-in-volume estimate or an explicit obstruction;
5. determine whether any action-window convolution has a useful asymptotic factorization;
6. extend the orbit analysis beyond simply connected two-dimensional boxes or replace it with a dimension-robust invariant;
7. construct an infinite-volume measure if that research direction is scientifically warranted;
8. specify and test a continuum scaling independently of the finite results;
9. decide whether a separately authored faithful SIT discretization is warranted after the physical SIT action is frozen publicly; and
10. obtain specialist review of the gauge, measure, orbit, and probability arguments.

These open problems cap the claims but do not erase the finite results.

14. Conclusion

The written lattice system is best understood as an auxiliary finite model inspired by SIT rather than as a specification of physical SIT. Within that scope, it has a coherent and reproducible mathematical core: a valid gauge action, a structural restricted-sector orbit correspondence, an explicit action mismatch, fixed-magnitude scalar factorization, and quartic magnitude pinning. Version 2 preserves the original transfer research program while making its conclusion-bearing premise explicit and refutable. The transfer problem remains open. No continuum, thermodynamic, physical-SIT, HSIT, SIT / Riemann, Riemann Hypothesis, deposit, mirror, submission, or public action is implied by the local validation of this final-preprint successor.

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